



Microsoft Active Directory Training

When: Between 10th –12th April 2024

Where: Onsite

Conductor: CybExer Technologies

Contracting Authority: e-Governance Academy

Program Overview

Overview

This course dives deep into Microsoft's Active Directory Domain Services (AD DS), the central nervous system managing user access, permissions, and resources in a Windows network environment. This course combines classroom instruction with interactive hands-on labs. The participants will gain theoretical knowledge through lectures and demonstrations, followed by practical exercises where they can implement their understanding by managing users, groups, and policies in a simulated Active Directory environment. By the end of this comprehensive course, the participants will possess the skills and knowledge to confidently manage and secure Active Directory environment, ensuring efficient user access and a robust network infrastructure.

Goals

By the end of this course, participants will be able to:

- **Explain the core concepts of AD DS:** Understand the role of forests, domains, sites, domain controllers, and organizational units (OUs) in a hierarchical AD structure.
- **Manage User and Group Administration:** Confidently create, manage, and modify user accounts, groups, and group policies, ensuring proper access control and security.
- **Implement Group Policy Objects (GPOs):** Leverage Group Policy Objects (GPOs) to enforce security settings, software deployment, and user configurations across your network.
- **Secure Active Directory:** Understand common AD security threats and implement best practices for user authentication, password policies, and domain controller security.
- **Troubleshoot AD Issues:** Have the knowledge to diagnose and troubleshoot common AD problems, ensuring smooth network operation.
- **Scripting:** Understanding how to powershell works and how to leverage it to administer standalone computers and also large active directory domains.
- **Desired state configuration:** How to leverage the desired state configuration to automate deployment, configuration and also security in windows environments and active directory domain services.

Requirements for Participants

Overview

The participants will use their own computers to participate in the trainings. All common workstation operating systems like Windows, Linux and macOS can be used. It is preferable that participants have full administrative access to their workstation operating system.

laptop minimum requirements for each participant:

- OS: MS Windows 10 x64 (or newer), GNU/Linux Ubuntu 18.04 x64 desktop (or newer) or macOS 11 (or newer)
- CPU: Intel i5 or equivalent (recommended – Intel i7 or equivalent)
- HDD: 50 GB of free space (recommended SSD based storage)
- RAM: 8GB (recommended - 16GB)
- NIC: physical or dongle based RJ45 10/1000

Agenda

Day 1 (10.04.2024)

Time	Agenda Items
09:00 – 10:45	• Intro and environment familiarization • Overview of windows central management
10:45 – 11:00	Break
11:00 – 13:00	• Active directory and its components
13:00 – 13:45	Lunch
13:45 – 14:45	• Group policies
14:45 – 15:00	Break
15:00 – 16:00	• Windows scripting and WMI
16:00 – 17:00	Meeting with Kadri Uiboleht, Adviser at the Estonian Ministry of Defence
18:30	Welcome dinner with Estonian experts at Restaurant FOTOGRAFIKA

Day 2 (11.04.2024)

Time	Agenda Items
09:00 – 10:45	• Labs to set up Active Directory via console access • Joining non windows systems to AD for central authentication
10:45 – 11:00	Break
11:00 – 13:00	• Windows Events and event forwarding
13:00 – 13:45	Lunch
13:45 – 14:45	• Powershell scripting
14:45 – 15:00	Break
15:00 – 17:00	• Powershell local and remote system management

Day 3 (12.04.2024)

09:00 – 10:45	• Powershell labs
10:45 – 11:00	Break
11:00 – 13:00	• Introduction to Desired State Configuration
13:00 – 13:45	Lunch

13:45 – 14:45	• Labs to set up Active Directory via DSC
14:45 – 15:00	Break
15:00 – 16:00	Meeting with Uku Arold, Chief of Estonian Defense Forces' (EDF) strategic communications department

Training Conductor

- **Tanel Tetlov**

Tanel Tetlov is a penetration tester and a with a complex background from private sector, government/military and academia. His expertise includes mobile security. Currently, Tetlov is also a member of the Estonian Cyber Defence League. Tetlov has been a key red teamer at NATO CCDCOE cyber exercises and carried out numerous penetration testing projects at university, business and government settings. Tetlov has worked at various NATO organisations as a vulnerability assessor, penetration tester and technical researcher.